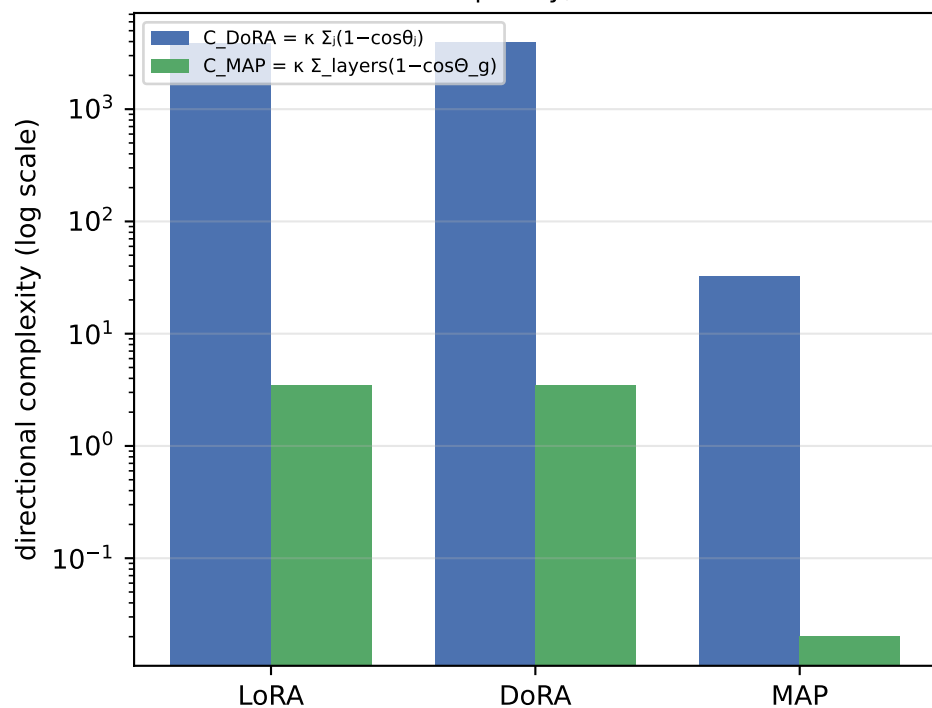


Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / MNLI

Fine-tuning metrics and aggregate directional complexity

Method	Eval acc	Trainable params	C DoRA (Σ local)	C MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	87.71%	1,920,003	3881.91	3.461	0.0833	0.0954	0.997
DoRA	87.74%	2,002,947	3911.60	3.500	0.0845	0.0960	1.000
MAP	85.13%	1,920,147	32.49	0.020	0.0051	0.0068	0.121

Local (sum) vs global (single rotation)
complexity, $\kappa=10$



Θ_{global} is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1 - \cos \Theta_{\text{global}} = \text{mean}_j (1 - \cos \theta_j) \rightarrow$ DoRA sums what MAP averages.

Per-column angle θ_j vs global angle Θ_{global}

